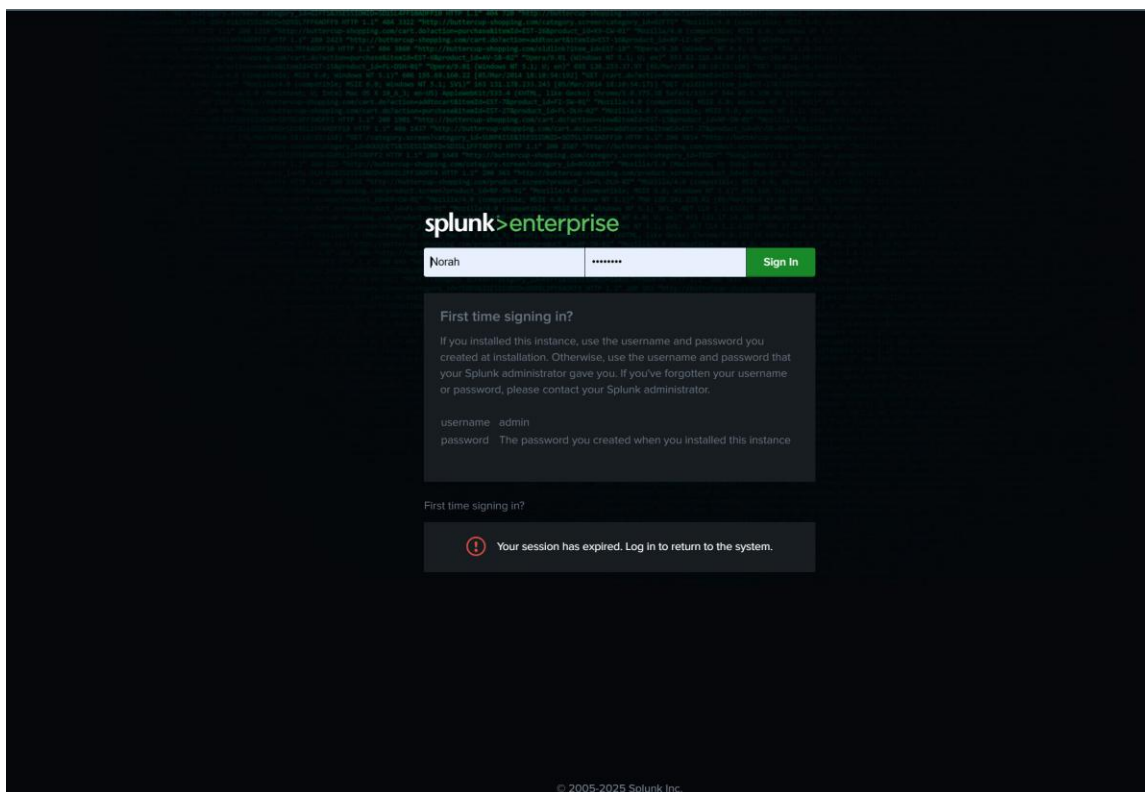


Phase 2: SIEM Dashboard Analysis (SSH Attack)

1. Splunk Installation & Access

- Installed Splunk Enterprise on a local machine.
- Accessed the Splunk Web interface via <http://127.0.0.1:8000>.
- Logged into the administrator panel to begin analysis.



2. Preparation

- Ensured access to the victim machine's SSH log file (auth.log) from Phase 1.
- Set up Splunk with an index named 'victim_logs' to receive log data.

3. Uploading Log Files

- Uploaded the Phase 1 SSH log file (auth.log).
- Assigned the index name 'victim_logs'.
- Set the source type to 'syslog' for consistent log parsing.

splunk-enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

What data do you want to send to the Splunk platform?

Follow guides for onboarding popular data sources



Cloud computing

Get your cloud computing data in to the Splunk platform.

10 data sources



Networking

Get your networking data in to the Splunk platform.

2 data sources



Operating System

Get your operating system data in to the Splunk platform.

1 data source



Security

Get your security data in to the Splunk platform.

3 data sources

4 data sources in total

Or get data in with the following methods



Upload

files from my computer

Local log files
Local structured files (e.g. CSV)
[Tutorial for adding data](#)



Monitor

files and ports on this Splunk platform instance

Files - HTTP - VM - TCP/UDP - Scripts
Modular inputs for external data sources



Forward

data from a Splunk forwarder

Files - TCP/UDP - Scripts

splunk-enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Next >

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. [Learn More](#)

Selected File: **auth.log**

Select File

Drop your data file here

The maximum file upload size is 500 Mb

 File Successfully Uploaded

FAQ

> What kinds of files can the Splunk platform index?

> What is a source?

> How do I get remote data onto my Splunk platform instance?

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Next >

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: auth.log

View Event Summary

Source type: Select Source Type

Save As

syslog

postfix_syslog
Output produced by the Postfix email server

sendmail_syslog
Output produced by the Sendmail email server

syslog
Output produced by many syslog daemons, as described in RFC3164 by the IETF

windows_snare_syslog
Output produced by the Snare syslog server on Windows

Format

Show: 20 Per Page

View: List

	Time	Event
1	4/28/25 7:01:01.000 AM	Apr 28 07:01:01 metasploitable sshd[1001]: Failed password for invalid user admin from 192.168.56.100 port 56789 ssh2
2	4/28/25 7:01:05.000 AM	Apr 28 07:01:05 metasploitable sshd[1002]: Failed password for invalid user root from 192.168.56.100 port 56790 ssh2
3	4/28/25 7:01:10.000 AM	Apr 28 07:01:10 metasploitable sshd[1003]: Failed password for vagrant from 192.168.56.100 port 56791 ssh2
4	4/28/25 7:01:15.000 AM	Apr 28 07:01:15 metasploitable sshd[1004]: Accepted password for vagrant from 192.168.56.100 port 56791 ssh2
5	4/28/25 7:01:20.000 AM	Apr 28 07:01:20 metasploitable sshd[1005]: Failed password for vagrant from 192.168.56.100 port 56792 ssh2
6	4/28/25 7:01:25.000 AM	Apr 28 07:01:25 metasploitable sshd[1006]: Failed password for vagrant from 192.168.56.100 port 56793 ssh2
7	4/28/25 7:01:30.000 AM	Apr 28 07:01:30 metasploitable sshd[1007]: Failed password for vagrant from 192.168.56.100 port 56794 ssh2
8	4/28/25 7:01:35.000 AM	Apr 28 07:01:35 metasploitable sshd[1008]: Accepted password for vagrant from 192.168.56.100 port 56795 ssh2

Add Data

Select Source Set Source Type Input Settings Review Done

< Back

Review >

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

- ☒ Constant value
☐ Regular expression on path
☐ Segment in path

Host field value

LAPTOP-2SPM3AQ6

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index

Default

[Create a new index](#)

- ✓ Default
- history
- main
- summary
- victim_logs

FAQ

- > How do indexes work?
- > How do I know when to create or use multiple indexes?

splunk>enterprise

Apps

Administrator

1 Messages

Settings

Activity

Help

Find

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< Back

Submit >

Review

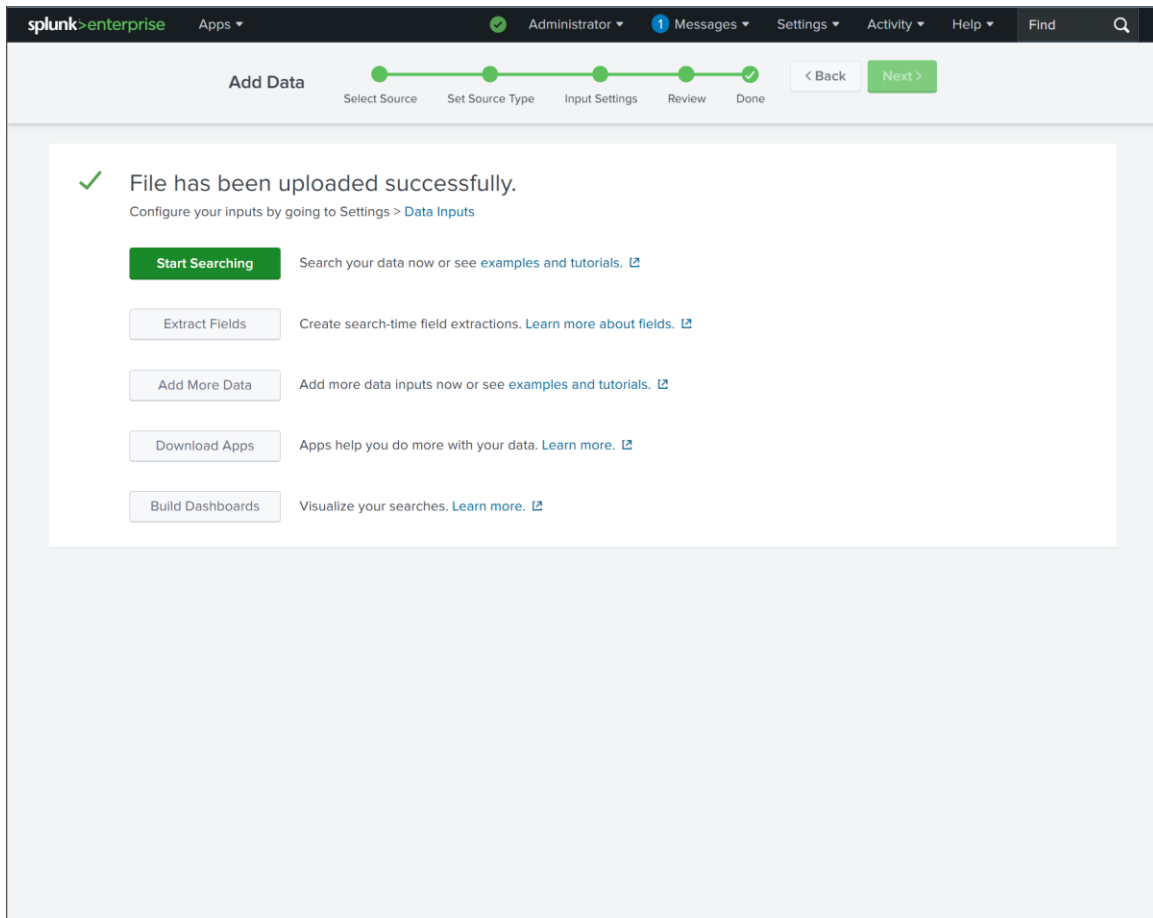
Input Type Uploaded File

File Name auth.log

Source Type syslog

Host LAPTOP-2SPM3AQ6

Index victim_logs



4. Visualizing Data in Splunk (Search & Reporting)

Initial Search Overview: Conducted a broad SPL search on the victim_logs index using sourcetype=syslog to confirm successful data ingestion and inspect raw event entries. This step verified that SSH-related log entries (e.g., "Failed password" and "Accepted password") were available for further analysis.

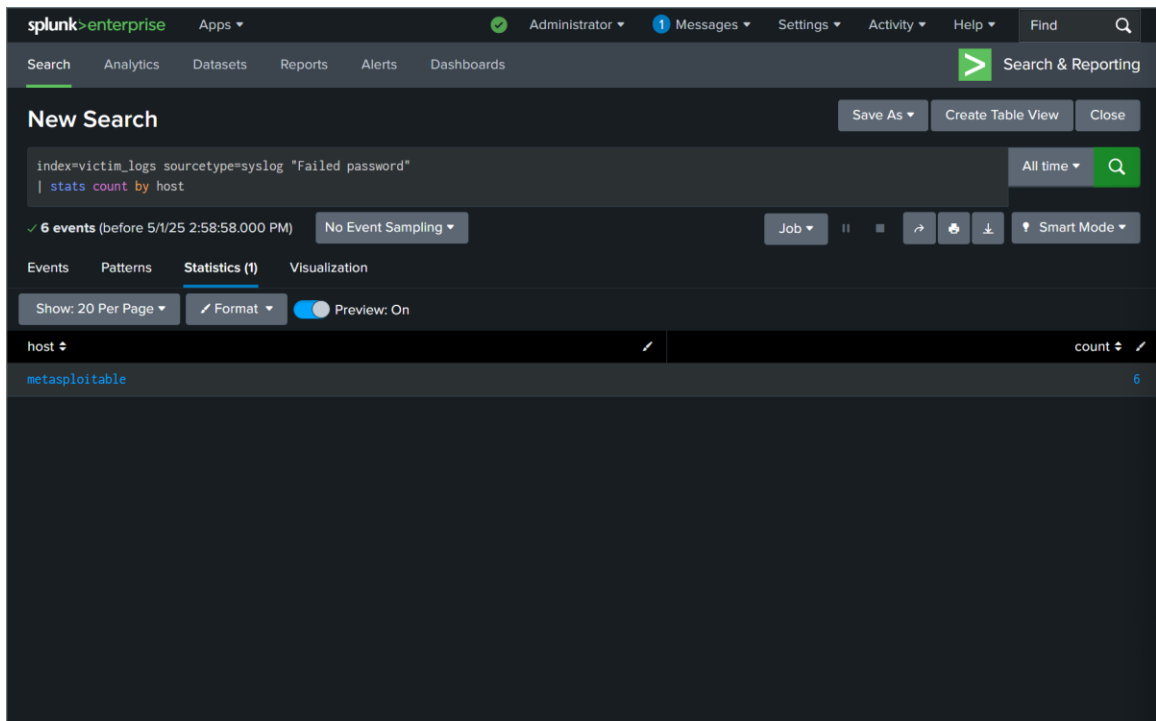
The screenshot shows the Splunk Enterprise web interface. At the top, there's a navigation bar with links to Search, Analytics, Datasets, Reports, Alerts, and Dashboards. Below this, the 'New Search' panel is active, displaying the search query `index=victim_logs sourcetype=syslog`. The results are shown in a table format, with columns for Time, _source, and _sourcetype. The table contains three entries for failed SSH logins on April 28, 2025, from the host 'metasploitable'.

Time	_source	_sourcetype
4/28/25 7:01:35.000 AM	Apr 28 07:01:20 metasploitable sshd[1005]: Failed password for vagrant from 192.168.56.100 port 56792 ssh2	syslog
4/28/25 7:01:30.000 AM	Apr 28 07:01:15 metasploitable sshd[1004]: Accepted password for vagrant from 192.168.56.100 port 56791 ssh2	syslog
4/28/25 7:01:25.000 AM	Apr 28 07:01:10 metasploitable sshd[1003]: Failed password for vagrant from 192.168.56.100 port 56791 ssh2	syslog

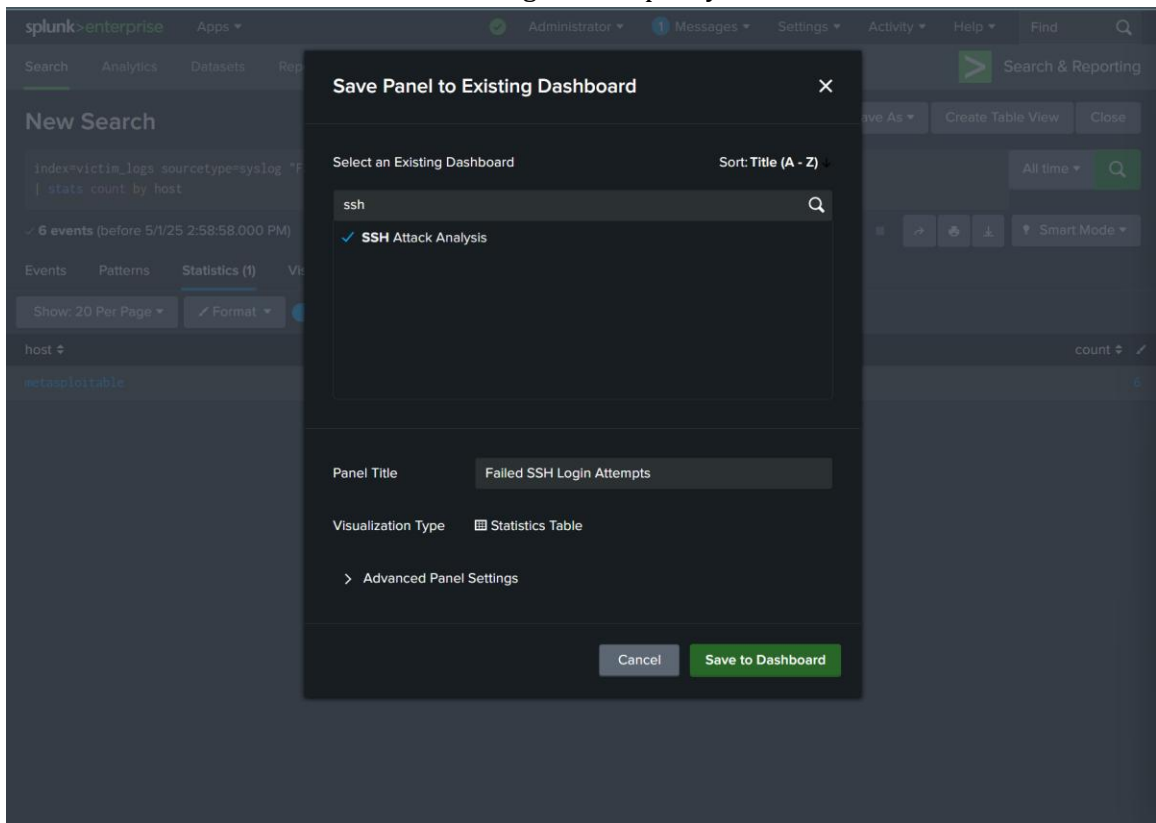
First Query – Failed SSH Logins

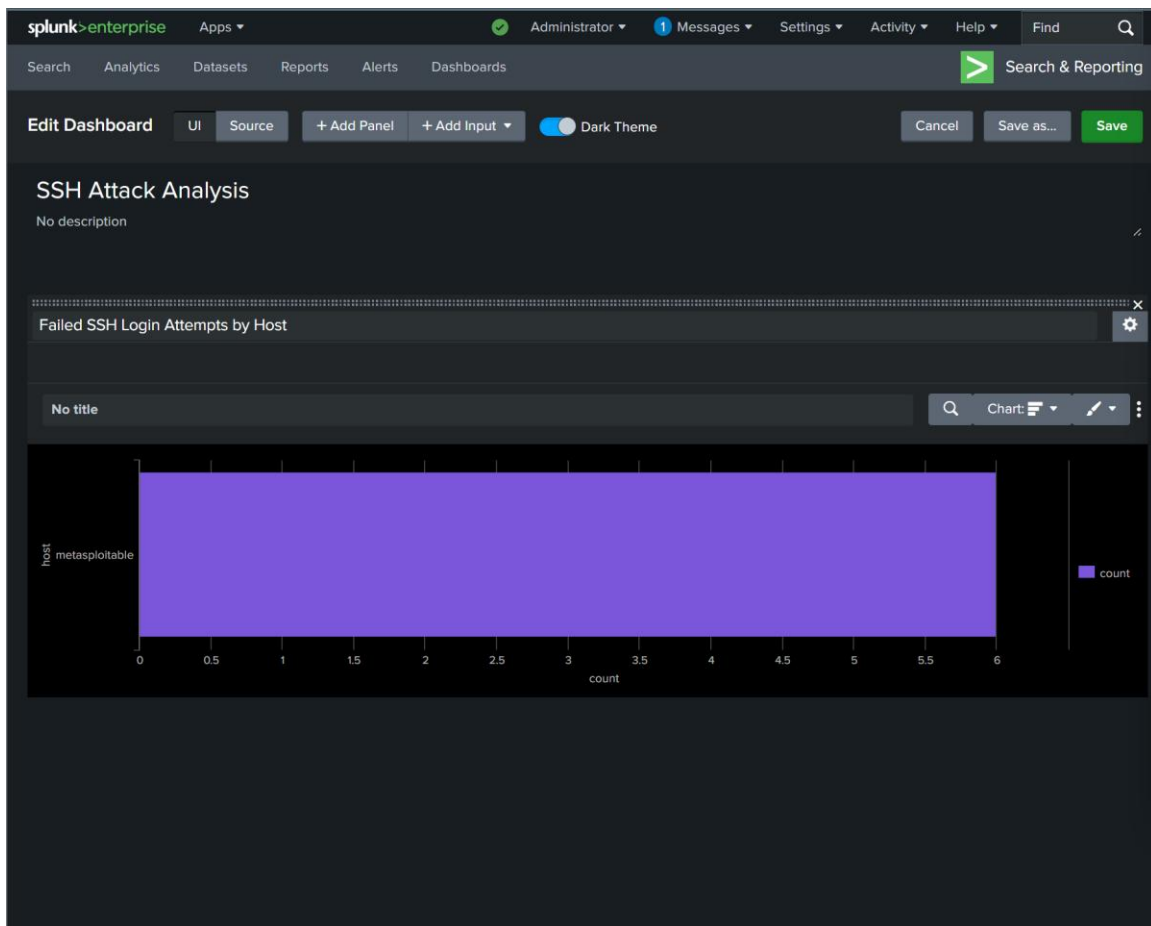
SPL:

`index=victim_logs sourcetype=syslog "Failed password"`
`/ stats count by host`



- Created a bar chart titled: Failed SSH Login Attempts by Host

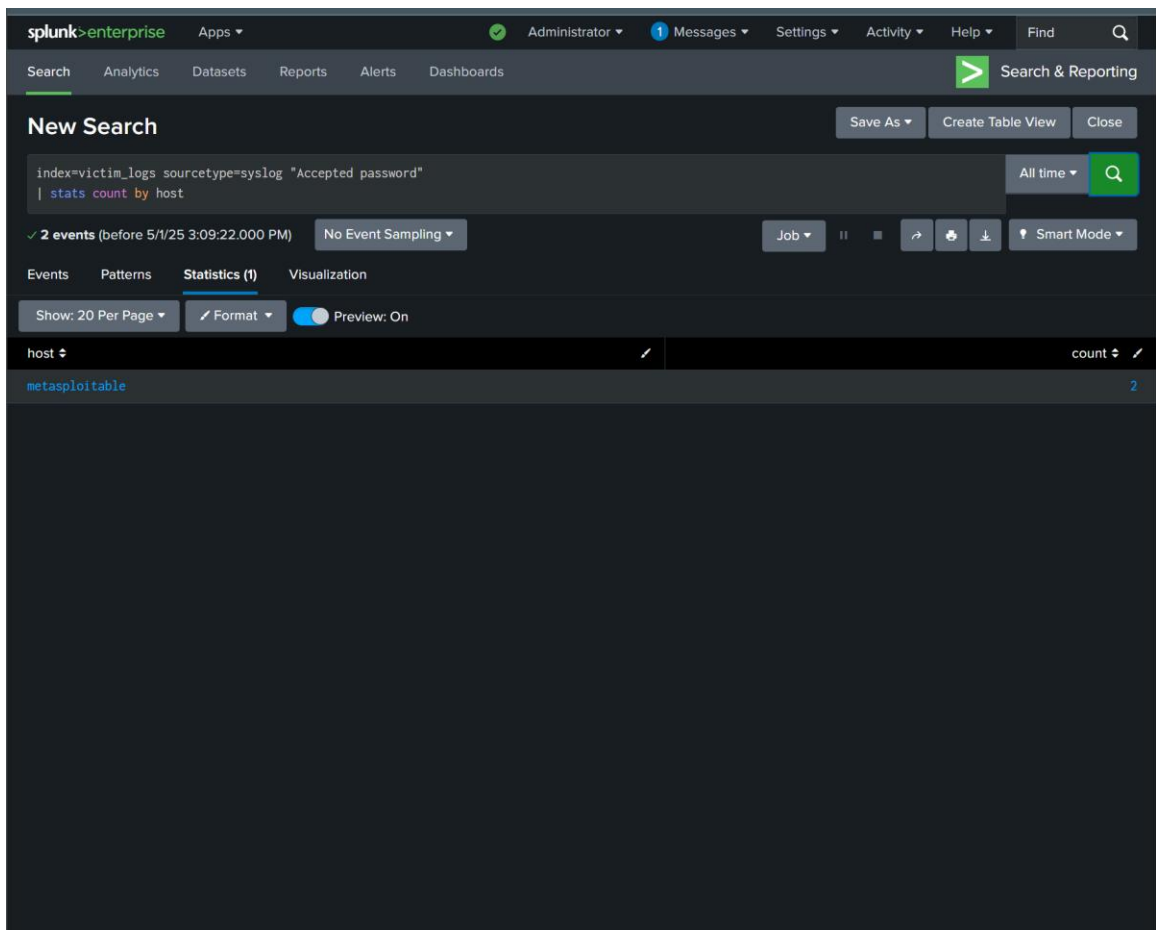




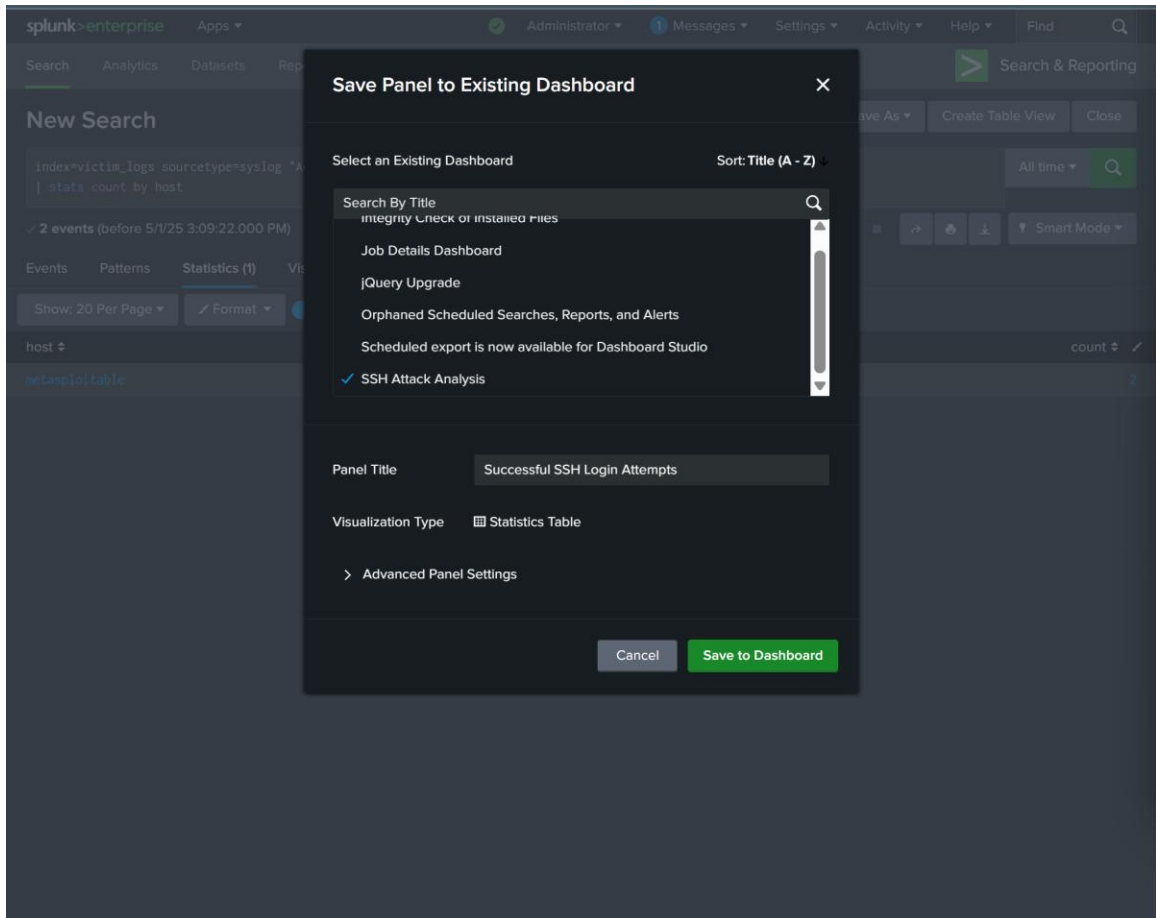
Second Query – Successful SSH Logins

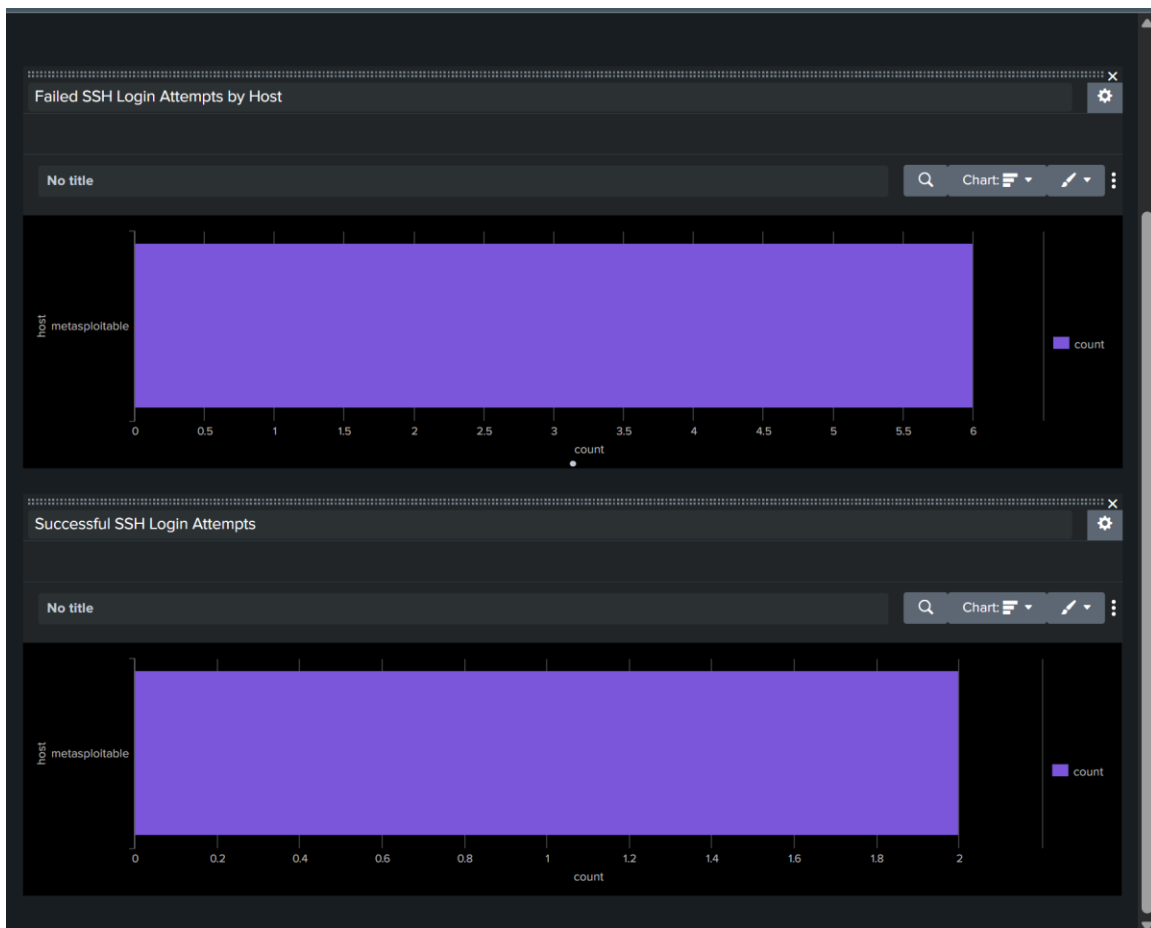
SPL:

index=victim_logs sourcetype=syslog "Accepted password"
/ stats count by host



- Created a bar chart titled: Successful SSH Login Attempts by Host

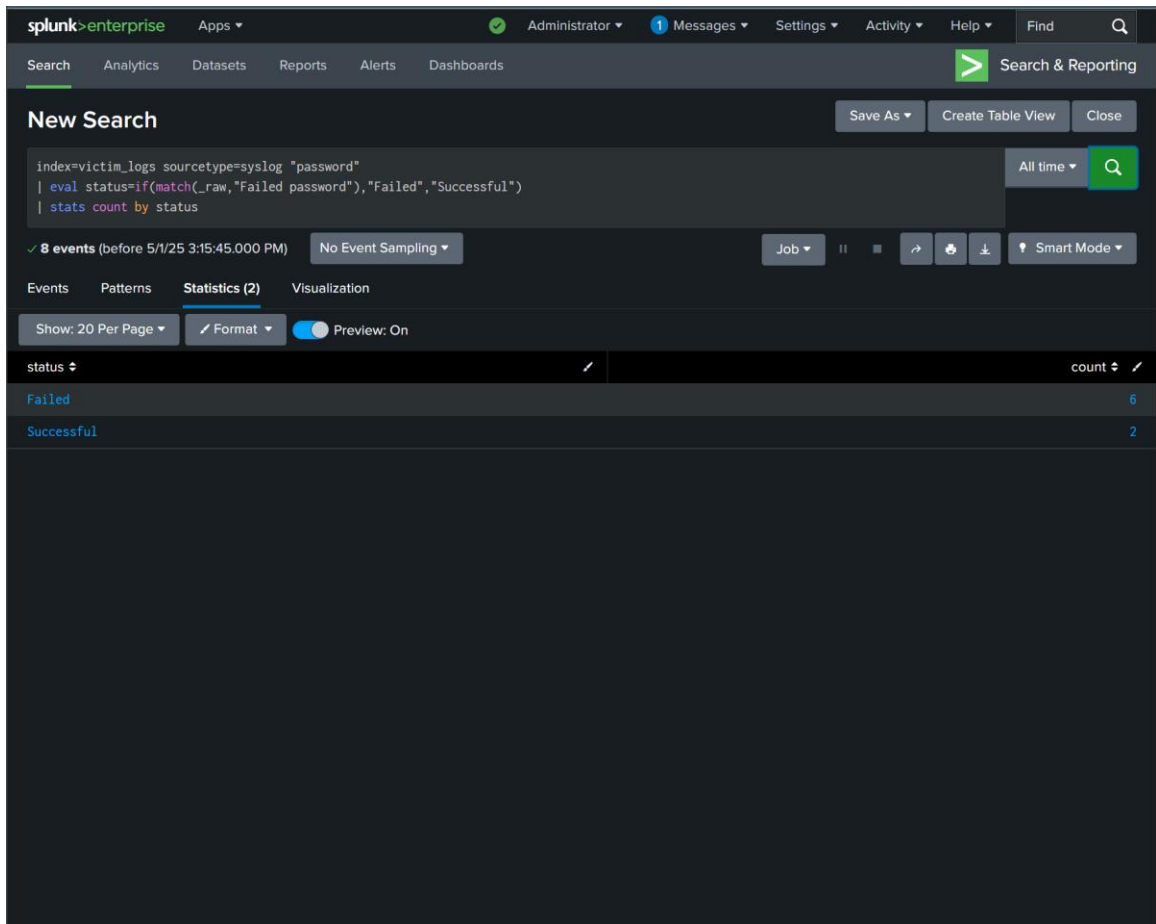




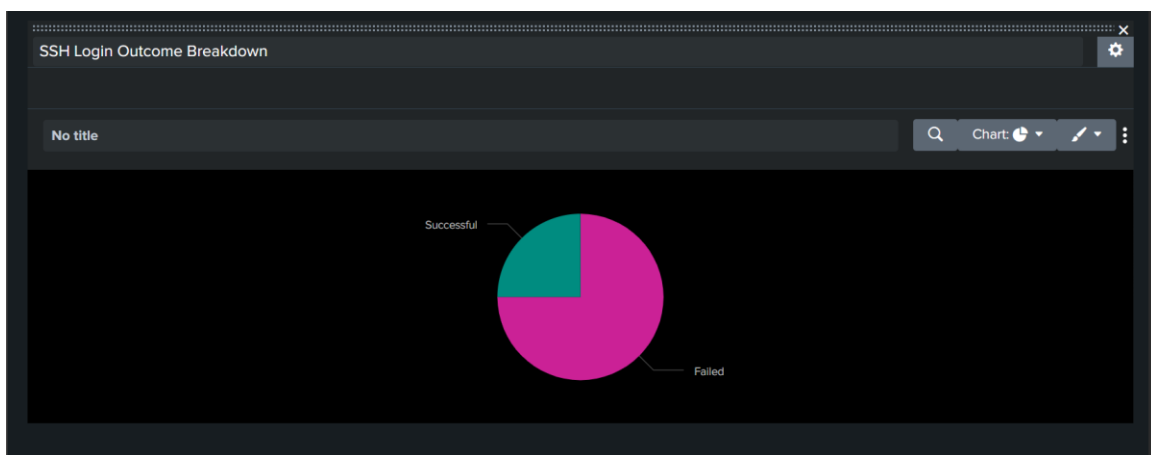
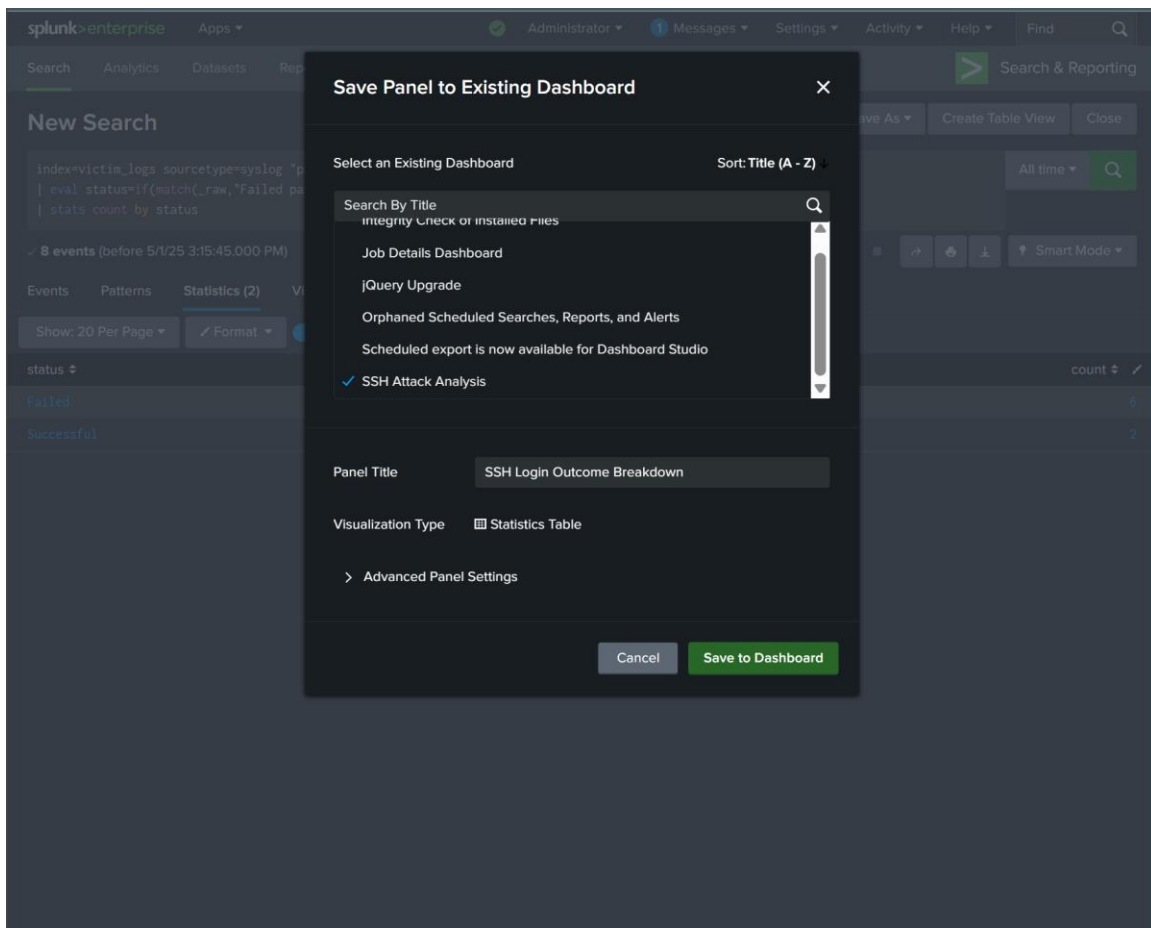
Third Query – Combined Outcome Breakdown

SPL:

```
index=victim_logs sourcetype=syslog "password"  
/ eval status=if(match(_raw,"Failed  
password"),"Failed","Successful")  
/ stats count by status
```



- Created a pie chart titled: SSH Login Outcome Breakdown

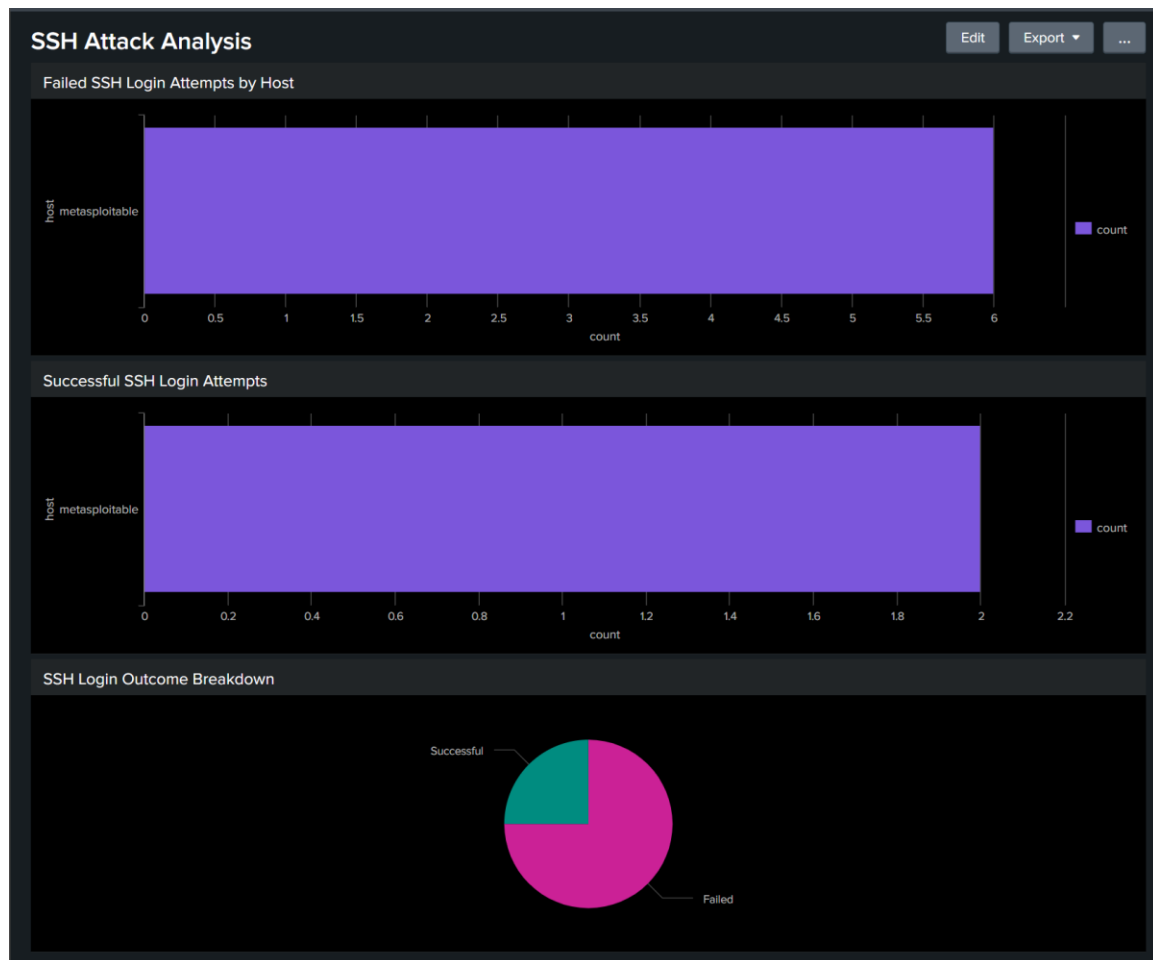


5. Final Dashboard – SSH Attack Analysis

Dashboard panels created:

1. Failed SSH Login Attempts by Host (bar chart)
2. Successful SSH Login Attempts by Host (bar chart)

3. SSH Login Outcome Breakdown (pie chart)



6. Analysis Outcomes

1. Splunk successfully ingested and parsed the victim machine's log file.
2. SSH authentication activity was clearly visualized, showing a higher frequency of Failed password events.
3. Attack data was effectively categorized into success and failure using SPL logic.
4. Visual dashboards highlighted brute-force patterns that could help detect SSH-based attacks.
5. The clear visuals supported quick decision-making for potential incident response.

7. Summary of Screenshots

Screenshot Title	Purpose
Splunk Installation Confirmation	Proves successful installation and access to the web UI at http://localhost:8000 .
Index Creation – victim_logs	Confirms the custom index used for log ingestion.
Log Upload – auth.log	Verifies log file was successfully uploaded and indexed.
Search Result – Failed SSH Logins	Demonstrates correct SPL query returned real failed login events.
Search Result – Successful SSH Logins	Shows the successful login entries and their SPL query.
Search Result – Login Outcome Breakdown	Illustrates the use of eval and status classification in SPL.
Dashboard – Full View	Displays all created panels together in one dashboard.
Panel – Failed SSH Login Attempts	Highlights the visualization of failed login trends.
Panel – Successful SSH Login Attempts	Visualizes accepted SSH login events.
Panel – SSH Login Outcome Breakdown	Summarizes login outcomes in a pie chart for quick insight.